

Remote Control for Rural Irrigation

Rural irrigation and farmers stand to benefit greatly from this innovative product, so let us see what it is

SNEHA A.

A water pump used for farmland irrigation is one of the key factors in the rural areas that need attention. The farmers have to walk long distances via rough and uneven terrains to reach the water pumps to turn them on/off. While doing this, they often encounter wild animals and extreme weather conditions. Also, they are unable to identify shock hazards due to wet areas and leakage of electricity and get themselves hurt.

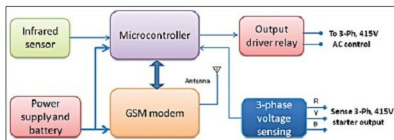
The above factors force farmers to keep their water motor pumps turned on continuously. To help farmers overcome this issue, and save wastage of water and electricity, and to control the pumps remotely, Ossian Agro Automation has developed an innovative product called Nano Ganesh-GSM.

Inspiration

Santosh Ostwal, CEO and founder director of Ossian Agro Automation Pvt Ltd, belongs to a farmer's family. An incident in the early years of his childhood impacted his life so much that it resulted in this innovative development.

He says, "When I was a kid, I witnessed my 81-year-old grandfather walking towards the water pump on a daily basis. His right leg was amputated below the knee due to gangrene and he had to use a support stick to walk. Despite this, he walked approximately one kilometre to the farm at midnight, just to switch on/off the pump. I used to accompany him. When I asked him the reason, the answer was very simple—power was available at night time only and we needed to irrigate half-dried oranges."

When Ostwal grew up and became an engineer, he was able to relate the



Block diagram of Nano Ganesh-GSM

requirement of the rural people and thought of using technology for the development of e-irrigation in India.

Hardware

Although Nano Ganesh-GSM comprises a number of blocks that play an important role, only the GSM module and microcontroller are visible, in addition to output stage, with all other hardware components simulated.

GSM modem. Different brands of GSM modules have been tried. The first version of Nano Ganesh used mobile phone for wireless connectivity, which has been replaced by a GSM module. This was a step to move towards M2M revolution in the field of mobile remote control.

SMPS power supply with battery back-up. Power requirement conditions for sensitive electronics goods are quite stringent. Care has been taken to ensure that the system works satisfactorily in extremely low, high and imbalanced voltage conditions as well. An external battery with charger is provided to allow easy replacement after two to three years.

Microcontroller and allied circuitry. Microcontroller being an intelligent device, it can help anyone easily optimise the system at very low cost. It intelligently replaces hardware logics built around diodes and transistors with simple software simulations.

Tone generator. Tone generator is

switching logics.

Driver stage and relay with isolation circuits. Driving the output relay requires special skills as it is really important to prevent the resetting of the microcontroller circuit when the system is doing a high-power application. Optical isolator circuits have been built around standard components available in the market.

Output sensing stage. It is important for the farmer to understand whether the pump is really on or off after sending the 'on' command. Many systems in the market provide a feedback from the motor-on relay as soon as it is activated by the microcontroller command. This is a virtual acknowledgment before the pump starts.

There is a possibility that due to low voltage or single phasing, auto switch does not start the pump and the farmer still gets the motor-on feedback. Nano Ganesh-GSM actually senses the output voltage of the starter and converts the signal into a digital form with the help of voltage converters. Hence, the farmer gets the motor-on SMS or call only after the pump has been actually started and has picked up speed.

Infrared sensing and processing stage. It uses a special infrared diode circuitry to sense the opening of the panel door and physical movement of a person standing in front of the panel. This allows the farmer to get an alert call when an unauthorised person

also a low-cost astable mode clock wave generator built around IC 555. It is effectively used to deliver different tones using a single IC and multi-